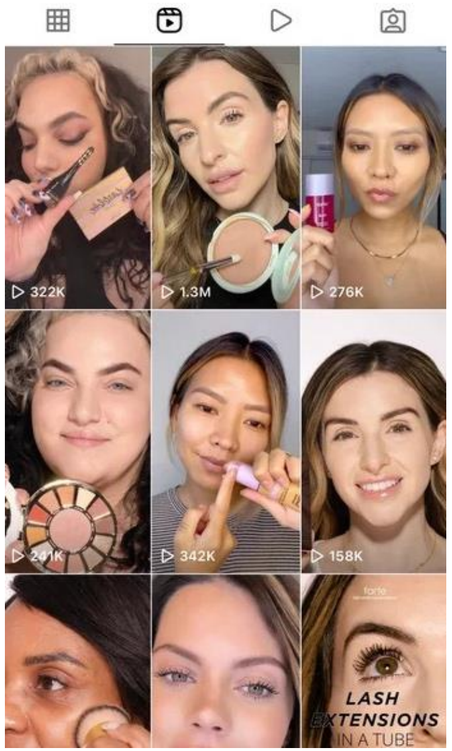
BEAUTY INFLUENCERS ARE OUR INSPIRATIONS



We can't give them advice with our "beauty" expertise...

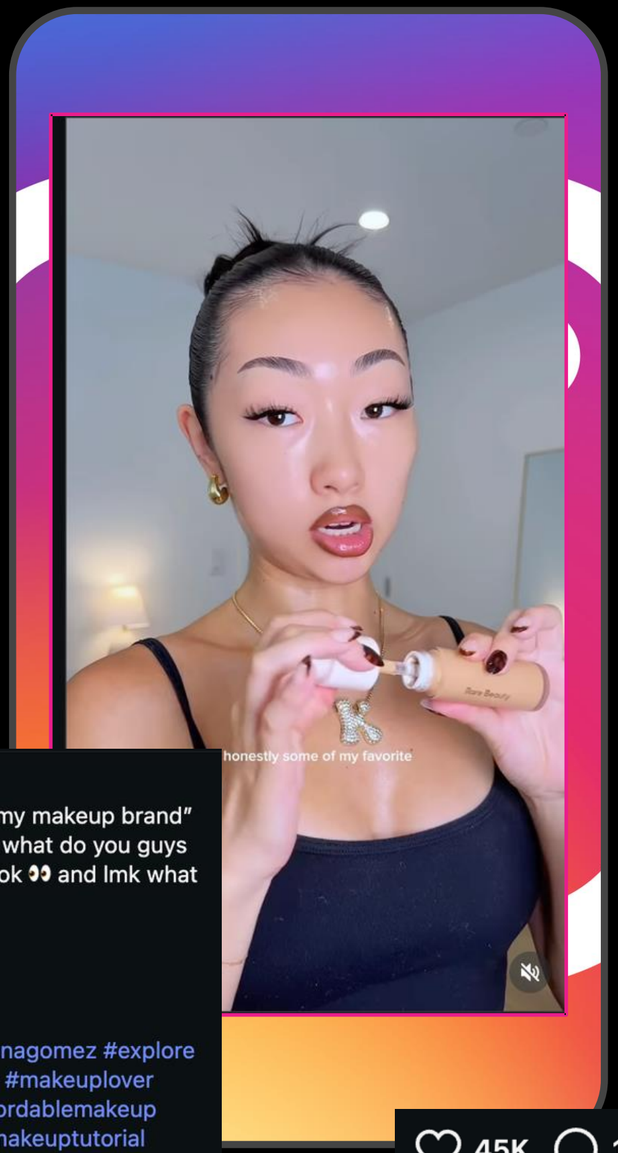
BUT WE CAN HELP INSTAGRAM BEAUTY INFLUENCERS WITH OUR ANALYTICS SKILLS!

WHAT MAKES A BEAUTY POST POP?

How to grow from Micro to Macro Instagram Beauty Influencer

by optimizing post engagement with data

9,821 instagram posts analyzed



itsbabykelz 21 w
sooo apparently "my makeup brand" is @rarebeauty 🤫 what do you guys think about this look 👀 and lmk what brand you get ✨
.
.
.
.
#rarebeauty #selenagomez #explore #makeup #beauty #makeuplover #makeupreel #affordablemakeup #makeuphacks #makeuptutorial #makeupideas #makeuplover #makeuplook #itgirl #makeuphaul #explorepage #viral #trending #makeupchallenge #makeuptips

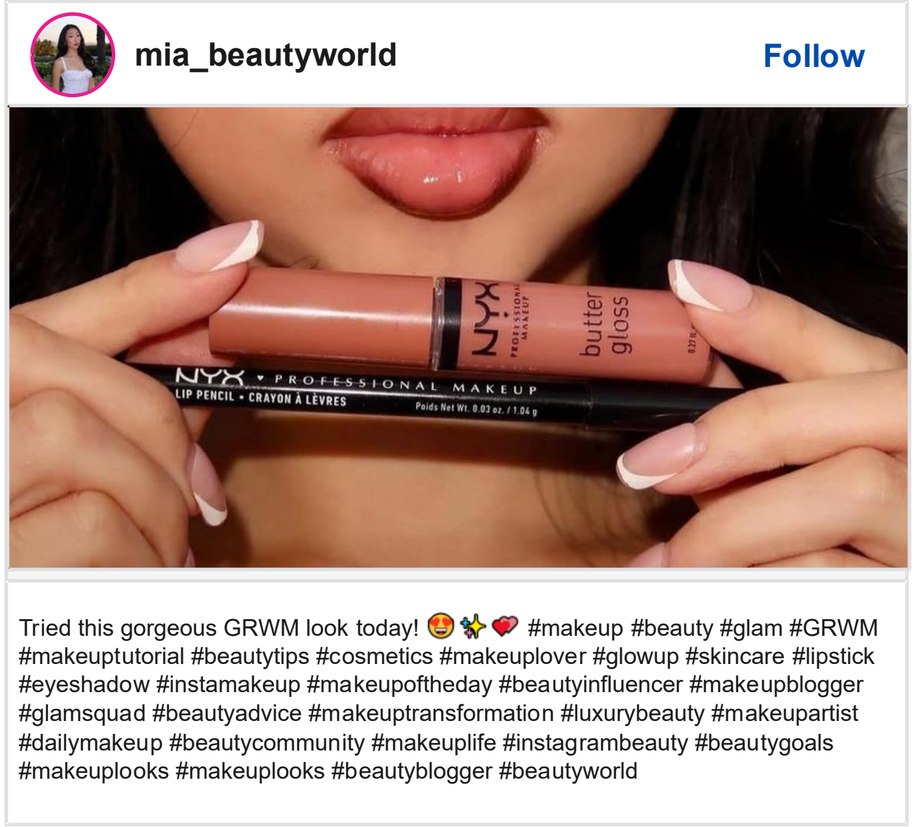
45K 182
3 November 2025

PRESENTED BY GROUP 8:
- JIAYI YANG
- NASRIN JALEEL
- NGOC DIEU LY VI
- NISA WU
- SHUAIYA XIONG

MEET OUR INFLUENCER

Meet Mia. She’s doing everything right.

Or so she thinks...



Her Post Performance:

Likes
312
out of 8,400 followers

Comments
14
people engaged

Engagement
3.88%
but it could be better

The Problem:

Mia has 30 hashtags, a polished photo and a solid caption, but she's still a **Micro Influencer.**

WHAT IS SHE MISSING?

9821 real Instagram posts from 1542 beauty influencers

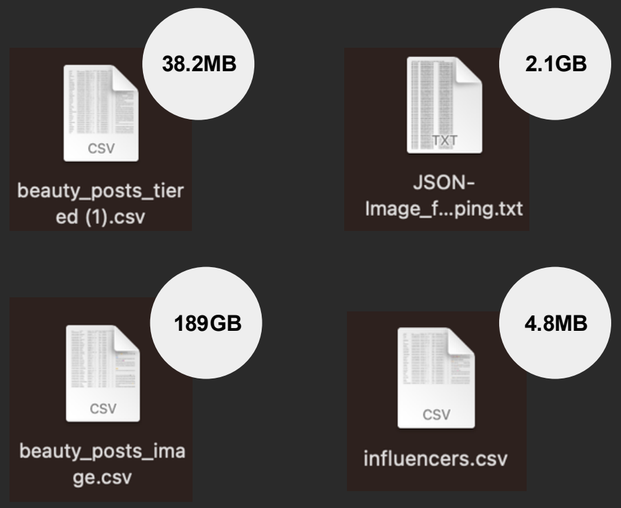
We asked the data.

9,821
Real Instagram Posts

1,542
Beauty Influencers

3
Tiers Analyzed

username	post_id	timestamp	likes	comments	followers	engagement_rate	caption
jedan711	1829231107826494086	1532281362	178	0	7878.0	0.0225945671490225	#polaroids #digitals #model #nyc #newyorkcity #c
ooh.mee	1924310888064755426	1543615755	11475	66	202206.0	0.0570754577015518	29.11.18 🤪
swalina	1998155492636976847	1552418718	60092	0	450119.0	0.1335024737902640	child of the moon 🌙 @marieschmidtcom
glam.glimmers	1952382315867152792	1546962130	22576	478	484968.0	0.0475371570907771	@elysian360 Wow all that hair 😍 Which one is yo
fays._	1791119100102367429	1527738057	35593	168	695865.0	0.0513907151530828	Bracitos de alambre, se tragaron una ciruela 💪 c Miau miau miau 🐱 #gymshark #gymsharkwomen #brazosdehilo #me
sofielindgren.se	1713319431930930664	1518463613	791	22	10099.0	0.0805030201010001	Bara jag som älskar fjädrar? ✨ #interiordesign#de
donialuvsu	1697299609776220804	1516553902	3322	51	226598.0	0.0148853917510304	Filming a Q&A on our marriage 1 year later ! Comi
kaysalbeautyxo	1886390934310327735	1539095344	125	0	2243.0	0.055728934462773	I think these lashes have become my favorite! 💖 Products used:



References: Kim et al. (2020) · Engagement Rate = (Likes + Comments) / Followers

From raw data to variables.

9 engineered features · 2 channels · 4 regression models

TEXT ANALYSIS

5 features

Text Cleaning

Lowercase · remove URLs, mentions · normalise whitespace

Effective Word Extraction

Language detection via langdetect · Google Translate API via deep-translator · Bag of Words · Differential word analysis

Sentiment Scoring using VADER

Lexicon-based NLP · compound score from -1 to +1 per caption

5 Variables Extracted

- Sentiment score
- Caption length
- Hashtag count
- Emoji count
- CTA flag

IMAGE ANALYSIS

4 features

Image Extraction

189GB archive · JSON-Image mapping · 7-Zip batch extraction · 9,821 images

Libraries - PIL + OpenCV

Per-image feature computation across all 9,821 extracted images

4 Variables Extracted

Brightness	Mean HSV value channel
Colorfulness	Hasler & Suesstrunk (2003)
Face Presence	OpenCV Haar cascade
Edge Density	Laplacian variance

References: Hutto & Gilbert (2014); Hasler & Suesstrunk (2003); De Veirman et al. (2017) · AI-assisted: Claude (Anthropic, 2025) used for feature engineering code.

Introduce the Tiers of Influencers.

MICRO

- **Under 20K followers - 5,021 posts**

Authentic, everyday creators

MID

- **20K to 100K - 2,921 posts**

Growing fast, starting to get brand deals, finding their niche

MACRO

- **Over 100K - 1,879 posts**

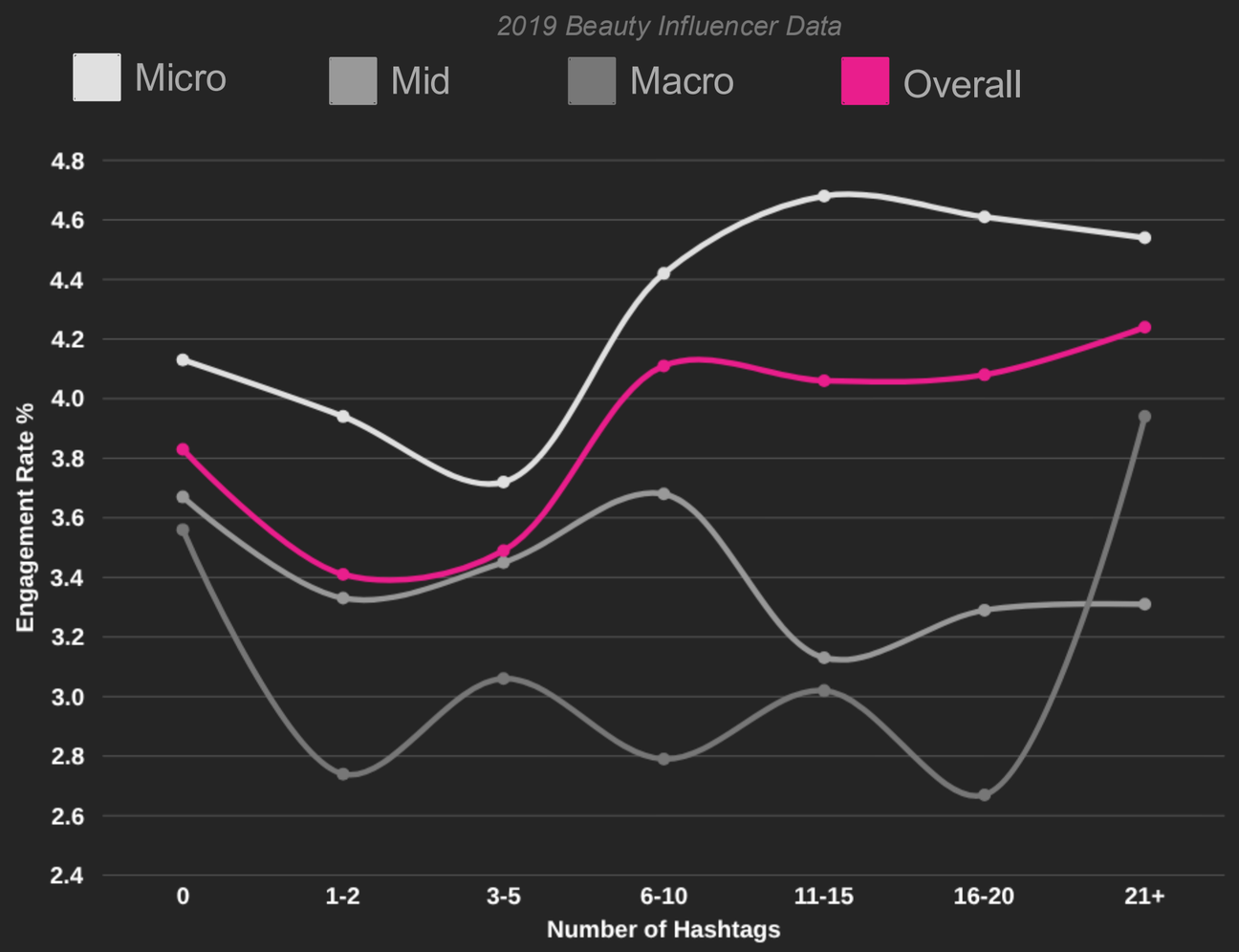
The ones with ring lights, photographers, and a PR team

References: Kim et al. (2020) · Engagement Rate = (Likes + Comments) / Followers

FINDING #1 · HASHTAG STRATEGY

Hashtag quality over quantity.

Instagram Just Made It Official.



References: Instagram (2025); Erz et al. (2018); Cheung et al. (2022); Socialinsider (2022) as cited in Hutchinson (2022).

WHAT OUR REGRESSION FOUND

Hashtag count coefficient = -0.0002**

Hashtag counts show a non-linear relationship with engagement rate.

- **Micro:** Clear sweet spot at 11-15, then plateau
- **Mid:** Relatively flat, modest peak at 6-10
- **Macro:** High baseline, but declines with more hashtags, organic reach does the heavy lifting

The Takeaway

Quality over quantity. Well-targeted hashtags outperform hashtag stuffing across all tiers.

Mia's fix: Swap 30 generic for 5 targeted hashtags

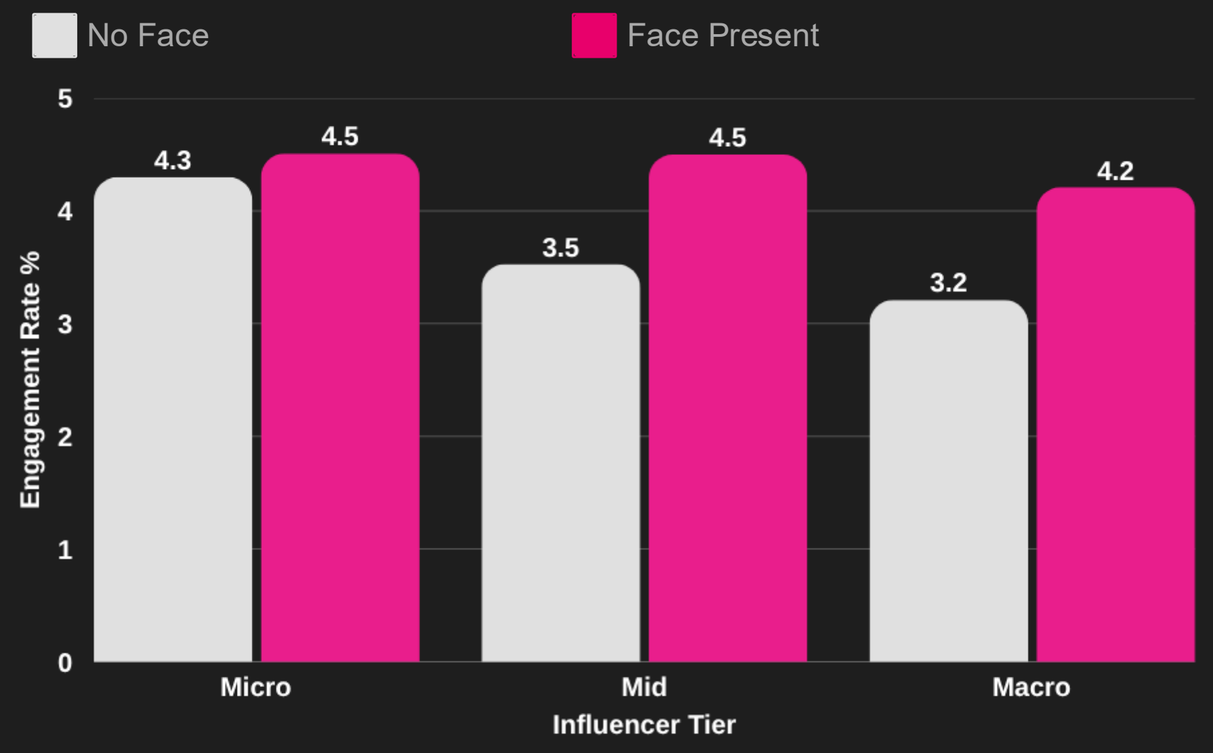
#GRWMtutorial #beautytransformation #makeuptips #kbeauty #beautycreator

*Socialinsider analysed over 75 million Instagram posts (Mar 2021 - Mar 2022) and found that **posts with 3-4 hashtags generated the highest average engagement rate** by impressions at 3.41%.*

Show Your Face.

Face in Post?

Mean engagement rate: face VS no face by tier



THE TAKEAWAY

Posts featuring a visible human face achieve higher engagement rates than product-only or scenery posts. The engagement premium from face presence grows as influencers scale.

Belanche et al. (2021) found that influencer credibility and para-social relationships on Instagram are built through personal, face-forward content. Tafesse & Wood (2021) found that influencer content strategy including personal imagery drives follower engagement.

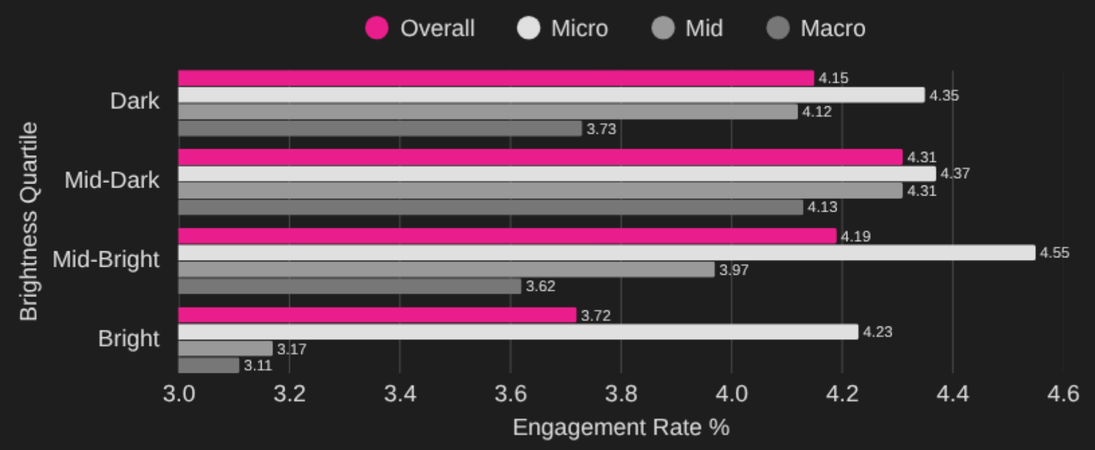
FINDING #3 · IMAGE QUALITY

Light It Right. Colourful but not saturated.

Two visual signals. Both statistically significant. Both easy to control.

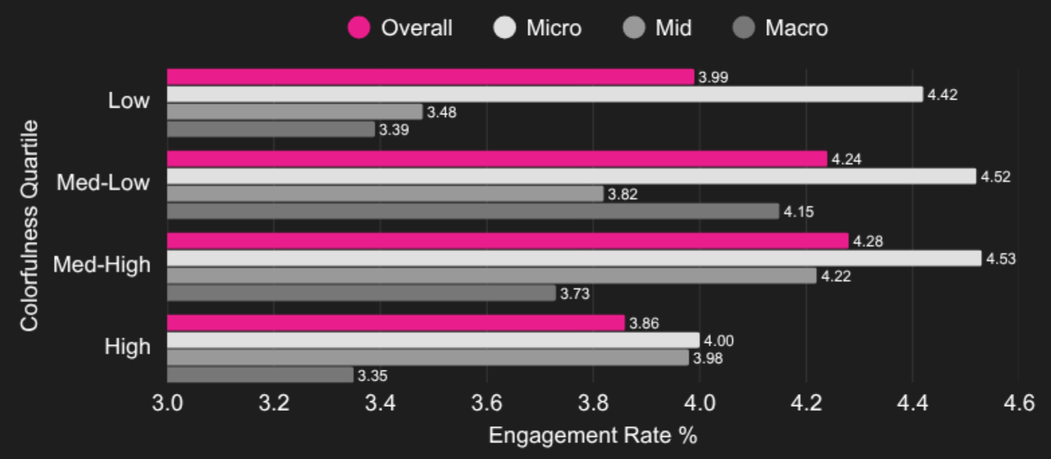
Brightness Level?

Engagement rate by brightness



Colourfulness?

Engagement rate by colourfulness



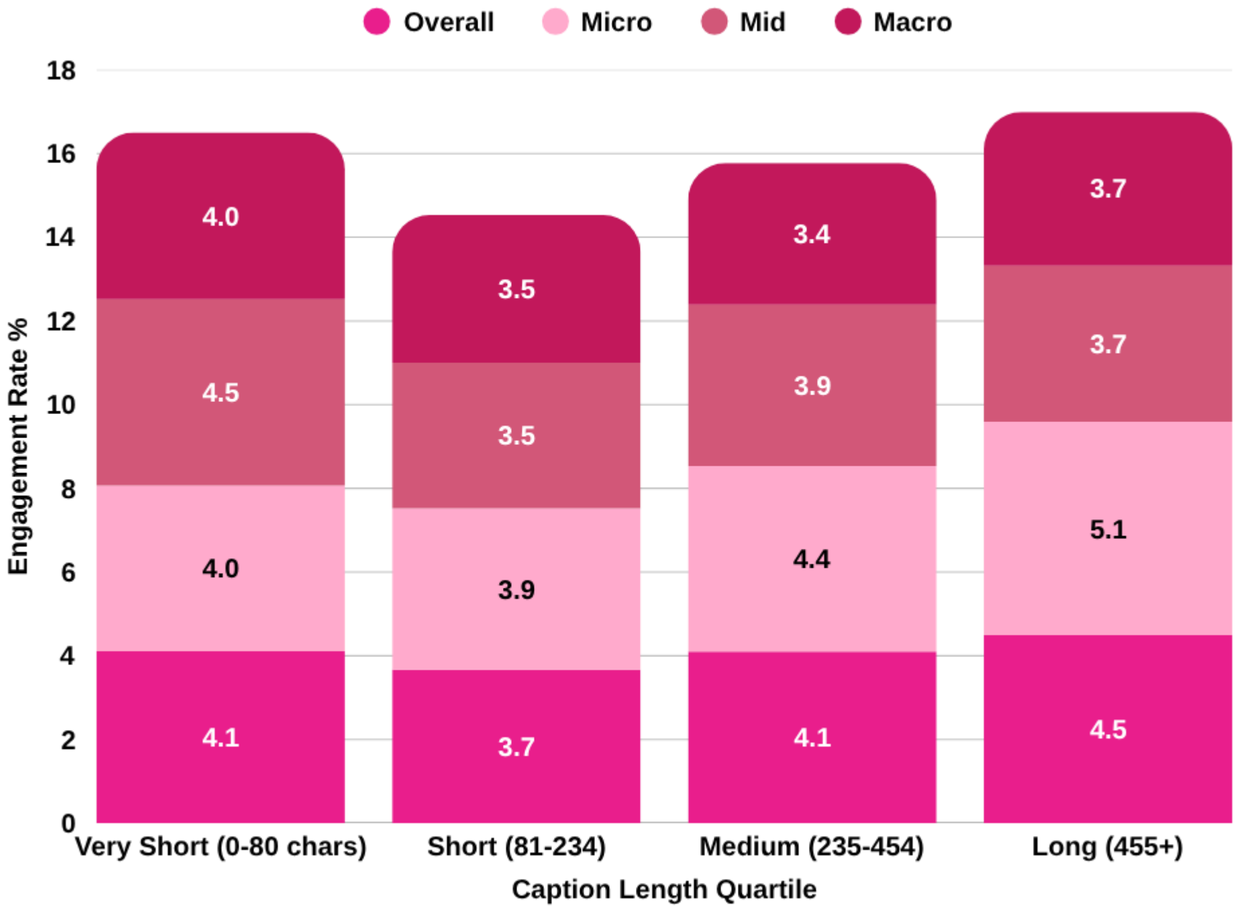
THE TAKEAWAY: 'natural but polished'

Mid-bright lighting and med-low colorfulness consistently outperform both extremes across all tiers. Over-filtered, oversaturated images signal inauthenticity. Audiences (and algorithms) reward images that feel real.

Mia's fix: Shoot in soft natural light, keep skin tones warm and true-to-life, and dial back the saturation.

References: Gelli et al. (2015). Image popularity prediction in social media using sentiment and context features. ACM Multimedia. AI-assisted: Claude (Anthropic, 2025) used for image feature extraction code.

Long Captions Win. Tone Matters Too.



Sentiment by Tier:

MICRO

Positive & upbeat tone performs best

MID

Neutral & raw tone outperforms positivity

MACRO

Bold, unfiltered opinions drive the highest engagement

Words that can boost captions:

MICRO

happiness · hydrating · base · cool · nice · recommend · believe · fine · ultra · weather

MID

steps · luck · flex · blossom · hype

MACRO

Hard to predict from words alone, audience size and brand authority dominate

Tafesse & Wood (2021) found that caption content strategy and tone are significant predictors of engagement. Cheung et al. (2022) found caption characteristics drive consumers' online brand-related activities.

Different Tier, Different Playbook.

Feature	MICRO	MID	MACRO
	< 20K followers	*20K - 100K followers*	*> 100K followers*
Hashtags	5 niche tags, max out the limit	3-5 mixed (niche + broad) tags	1-3 highly curated, or none
	** Instagram now limits all posts to 5 hashtags max (Dec 2025)*		
Face in Post	Always	Always	Critical
Caption Length	Long & chatty	Medium to long, narrative	Short to medium, punchy
Brightness	Medium bright	Medium dark	Medium dark
Colourfulness	Medium low, natural tones	Medium high	Medium low
Sentiment	Warm & positive	Neutral / raw let the story speak	Direct & unfiltered skip the positivity

References: De Veirman et al. (2017); Boerman (2020).

RECOMMENDATIONS

How to Go from Micro to Macro ?

01 Show Your Face

Always include your face.
It's worth ~+1pp in engagement
and it builds community.

02 Light It Right

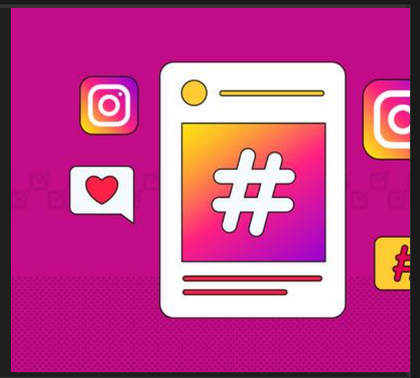
Medium brightness = maximum engagement
Overexposed = audience scrolls past

03 Write More

455+ characters outperform short captions.
Tell a story, add a call-to-action (CTA), be you.

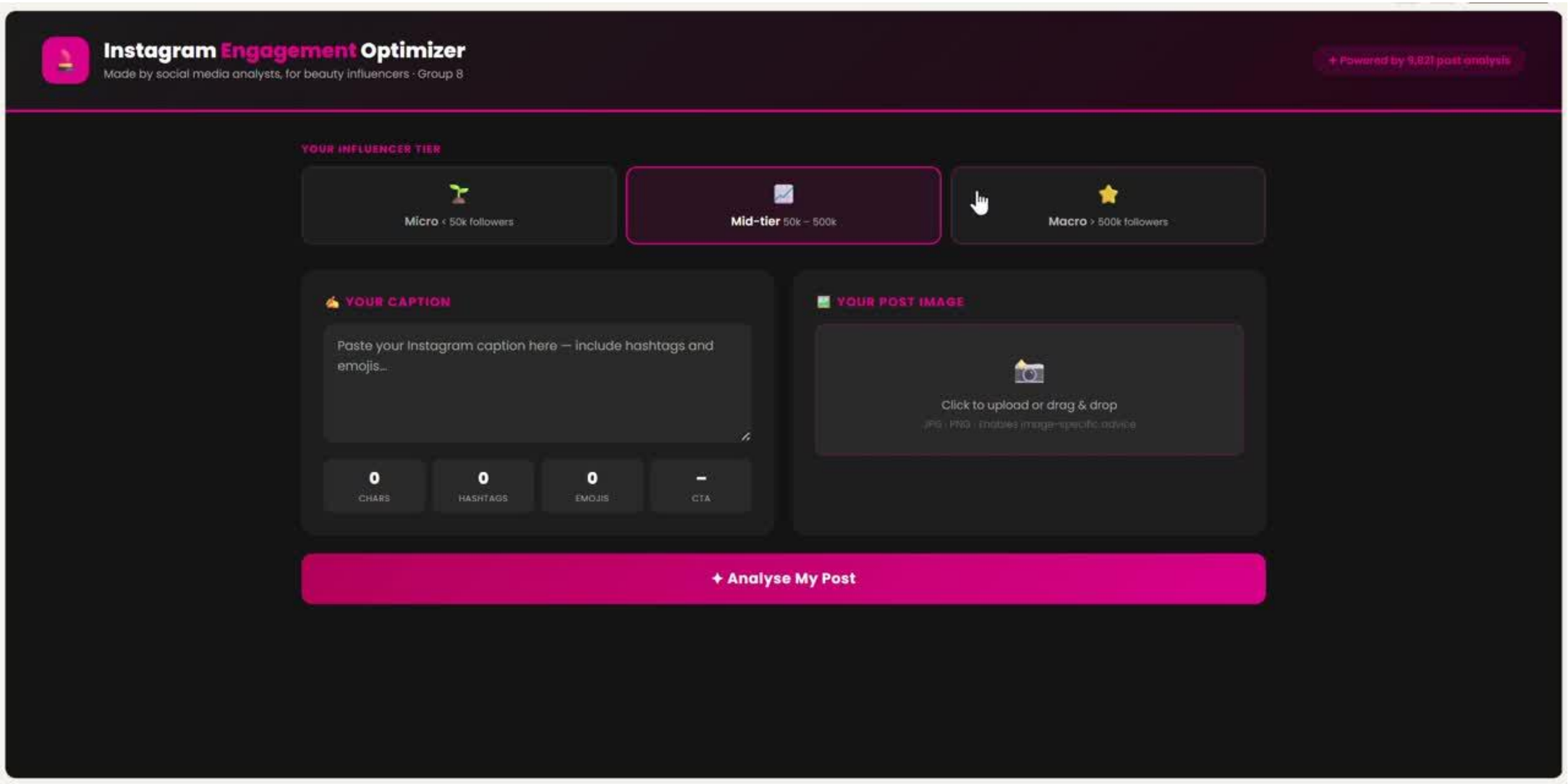
04 Tag Strategically

5 Relevant hashtags in every post.
Mix niche + broad for discoverability.



WE MAKE OUR RECOMMENDATIONS EASY TO FOLLOW:

INSTAGRAM ENGAGEMENT OPTIMIZER TOOL



Built with assistance from Claude (Anthropic, 2025).

MADE BY SOCIAL MEDIA ANALYSTS, FOR BEAUTY INFLUENCERS.

Thank You!

Q & A

GROUP 8

Appendix 1

https://github.com/nisawooo/social-media-group8

1. Model Description

Two OLS regression approaches were applied to predict post-level engagement rate (likes + comments/followers) across 9,737 Instagram beauty influencer posts.

- Models 1-3: Sequential OLS using caption features only, image features only, and all features combined, to compare channel-level explanatory power
- Model 4: Tier-stratified OLS re-run separately for micro, mid, and macro influencers

Caption predictors: Sentiment score, caption length, hashtag count, emoji count, CTA flag
Image predictors: Brightness, colorfulness, face presence, edge density

2. Intended Use

Designed to help beauty influencers and social media managers understand which content choices drive engagement, and how optimal strategies shift across audience tiers. Suitable for content planning and benchmarking, not for influencer contract decisions or campaign budget allocation.

Appendix 2

3.Benchmark different techniques

Model	Why we did not use it
Logistic regression	Our outcome (engagement rate) is continuous, not binary
Random forest / XGBoost	Black box — cannot extract directional coefficients for advice
LASSO / Ridge	Useful for high-dimensional feature selection — we only have 9 features, no regularisation needed
Fixed effects panel	Would require multiple posts per influencer over time — our data is cross-sectional
Poisson / negative binomial	Appropriate for count outcomes — engagement rate is continuous
Neural network	Prediction tool, not explanation tool — R^2 of 0.024 would not justify the complexity cost

Appendix 3

4. Fairness Considerations

- The dataset covers Instagram only; findings may not generalise to TikTok or YouTube
- Macro-tier subsample is smaller (n=1,854) and may produce less stable estimates
- Engagement rate is partly shaped by Instagram's algorithm, which the model cannot account for

5. Limitations

- R^2 values are low (0.018-0.033). Content features explain only a small share of engagement variance; posting time, trends, and algorithm state are not captured
- Correlation only face presence predicting higher engagement does not mean adding a face will causally increase engagement
- Cross-sectional design; it does not capture how strategies evolve over time

Appendix 4

6. References

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- Tafesse, W. and Wood, B.P. (2021) 'Followers' engagement with Instagram influencers: The role of influencers' content and engagement strategy', *Journal of Retailing and Consumer Services*, 58, p.102303. doi:10.1016/j.jretconser.2020.102303.

Appendix 5

Instagram Influencer Dataset External Inbox ×

Seungbae Kim <ksb2043@gmail.com> Mon 23 Mar, 14:06

to bcc: me

Hi,

Thank you for your interest in the Instagram Influencer Dataset.
Here is the link for the dataset.